

EcoPassEU: A Privacy-Aware Digital Product Passport DApp for Sustainable Consumer Goods in Europe

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1 Introduction

Every year, billions of products enter the European market with sustainability labels, yet consumers have little reliable way of checking whether claims such as “low carbon” or “ethically sourced” are accurate. A study commissioned by the European Commission found that more than half of the green environmental declarations lack verifiable evidence to support, eroding public confidence in corporate environmental commitments [22].

To address this credibility gap, the Ecodesign for Sustainable Products Regulation (ESPR) now mandates Digital Product Passports (DPPs) —standardised lifecycle records—for most EU products [8]. With the implementation of compliance in 2027, reliable DPP solutions have become a market urgent need[8].

Our product EcoPassEU is a blockchain-based DPP platform which allows manufacturers to record product information as permanent, publicly verifiable entries on the Ethereum blockchain. Unlike traditional databases, the immutability of blockchain records means that once the data is released, it cannot be altered retrospectively, eliminating the possibility of post-hoc falsification [6]. Any user can obtain a complete and encrypted product history by entering the product ID or scanning the QR code.

EcoPassEU serves four key stakeholders. It provides SME manufacturers with a cost-effective ESPR compliance channel through a web for batch data entry and automated QR code generation. At the same time, consumers, regulators and retailers can also get open and wallet-free access. Just search for the product ID to immediately obtain the blockchain-verified product passport and supply chain map, enabling the independent verification of environmental claims without prior technical expertise.

The value of EcoPassEU is threefold. Primarily, it provides manufacturers with a practical way to ESPR deadlines, reducing the risk of market exclusion. Meanwhile, the platform directly cracked down on greenwashing, securing product records via cryptographic hashing replaces brand self-reporting with independent, third-party verification. Thirdly, it promotes the development of circular economy by equipping recyclers with the precise material data necessary to recover resources when products are scrapped.

2 Background

2.1 Digital Product Passports and the Regulatory Context

A Digital Product Passport (DPP) is a structured digital record encompassing lifecycle data—such as material composition, provenance, and end-of-life guidance—designed to inform stakeholders and support more sustainable design choices, efficient resource use, and the transition towards circular business models [1,2,5,15].

The regulatory foundation for DPPs comes from the European Green Deal and Circular Economy Action Plan [11,12]. Based on these 2 frameworks, the Ecodesign for Sustainable Products Regulation (ESPR), came into force on 18 July 2024,, mandates DPPs for almost all physical goods sold in the EU, regardless of the seller’s location [8].

2.2 The Problem: Trust in Sustainability Claims

Despite regulatory momentum, the implementations of most DPPs rely on centralised, proprietary databases [20]. This architecture encourages the selective concealment of data, hindering aftermarket participants like independent repairers [9]. Furthermore, managing huge volume of lifecycle data centrally raises significant concerns regarding storage efficiency, data integrity, and vulnerability to breaches [26]. Taken together, downstream actors — including importers, recyclers, and end consumers — lack reliable ways to independently verify product information.

This problem is exacerbated by the prevalence of greenwashing, broadly understood as false, vague, or unsubstantiated environmental claims that make products appear more sustainable than they are [16]. A 2020–2021 EU sweep of 344 online claims found that nearly half were potentially deceptive—often substituting verifiable details with vague terms like “conscious”—and 59% lacked accessible evidence [22]. Another EU website screening found that 42% of sustainability claims were misleading enough to constitute improper commercial behaviour [22].

Blockchain provides a structural solution to this trust problem. Because records on a public blockchain are replicated across a distributed network, neither single party can unilaterally tamper with the records, providing a form of tamper evidence that cannot be replicated by a centralised system [6,19]. Smart contract introduced in the Ethereum protocol allows data access and update rules to be encoded in publicly readable logic, so that there is no need for trusted intermediaries and enabling cross-organisational verification without shared infrastructure [28].

2.3 Related Work and Existing Platforms

Blockchain-based product traceability uses its immutability and decentralised verification to improve supply chain transparency [6]. Blockchain-based product traceability emerged as a research topic around 2015–2016, with early work

concentrated in food and agriculture before spreading to pharmaceuticals, electronics, and textiles [10,23]. More recently, research has shifted from feasibility towards scale and efficiency: for instance, Walmart’s use of IBM Food Trust — built on Hyperledger Fabric — to reduce food-item tracing time from six days to under three seconds [10] .

In the commercial field, Circularise is among the closest comparable platforms, offering DPPs with selective disclosure of sustainability data across supply chains, with applications spanning the automotive, battery, and textile sectors [7]. VeChain similarly deploys blockchain and smart contracts on its VeChainThor platform to achieve supply chain transparency, though it runs on a permissioned network with a restricted set of authority-approved validators [25]. Both platforms have a common limitation: reliance on proprietary infrastructure or permissioned networks introduces vendor dependency and reduces interoperability [3,27].

EcoPassEU takes a different approach. Rather than relying on a proprietary database, it is built on the public Ethereum blockchain and IPFS, with all rules encoded in open smart contracts. Product data is stored on IPFS, whilst only a small set of cryptographic identifiers is written on-chain, keeping costs low. The batch registration function further reduces the cost per product, making the platform viable for SMEs. And as an open-source project, it carries no vendor lock-in risk.

2.4 Business Viability

EcoPassEU’s revenue model rests on two foundations. The first is public funding: the EU’s CIRPASS-2 project (2024–2027) explicitly subsidise DPP deployment across textiles, electronics, and construction, with dedicated support for SMEs adopting “DPP-as-a-Service” models [13]. As an open source platform, EcoPassEU is fully capable of benefitting from such grants.

Another is subscription revenue. ESPR compliance is legally mandatory for any business selling physical goods in the EU, forming a captive market independent of voluntary commitments [8]. The green technology and sustainability market is projected to grow from USD 25.47 billion in 2025 to USD 73.90 billion by 2030 [18]. EcoPassEU charges a graded subscription fee based on passport volume, with optional premium tiers for API integration and compliance reporting. The platform is built on public Ethereum, supporting payment with ETH or other ERC-20 tokens, providing international users a lower-friction settlement option and removing the need for traditional banking intermediaries.

3 Design Documentation

3.1 Design Philosophy and Architectural Choices

EcoPassEU is a DPP designed to deliver ESPR-compliant Digital Product Passports with independently verifiable sustainability data. The design is governed by three design principles, each addressing a specific product requirement.

First, a hybrid storage model is adopted: full product data is pinned to IPFS via Pinata [4], whilst only the content identifier (CID), a `keccak256` integrity hash, the issuer address, and a timestamp are recorded on-chain [28], keeping gas costs low. Second, commercially sensitive data remains off-chain by design; the on-chain hash serves as an integrity anchor, confirming that the off-chain document is unaltered without exposing its contents to the ledger. Third, all read queries use a public RPC endpoint, allowing consumers and regulators to inspect any passport without holding a wallet or incurring gas fees.

3.2 System Architecture

Figure 1 shows the four-layer architecture of the overall system

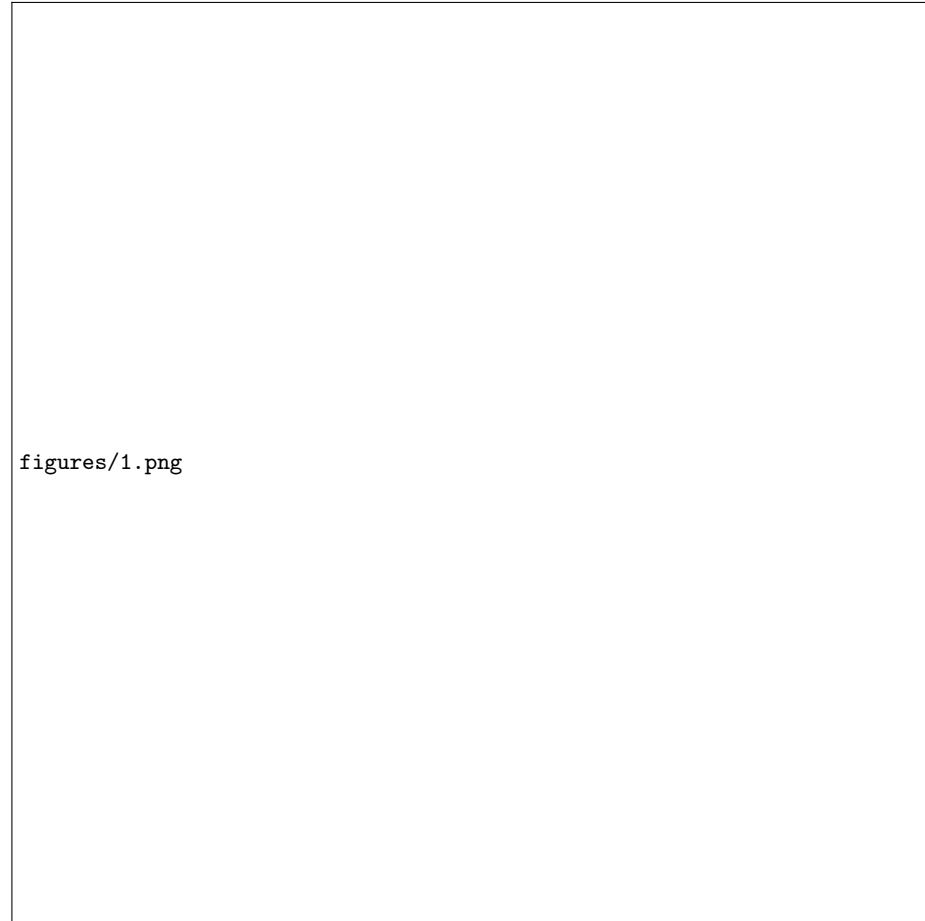


Fig. 1. EcoPassEU system architecture.

At the user layer, two distinct actors participants interact with the system: manufacturers, who connect through MetaMask to sign and submit transactions, and consumers or regulators, who can query product records without a wallet. Both routes pass through the application layer — a React/Vitefrontend deployed on Vercel, using Ethers.js v6 for contract calls — which is responsible for uploading documents to IPFS, calling the Sepolia test network and local integrity check. The `PassportRegistry` smart contract records the CID, metadata hash, issuer address, and timestamp on the Sepolia ledger, emitting a `PassportIssued` event upon registration. An `exists` flag can prevent duplicate entries. Supply chain provenance is visualised via Leaflet.js with OpenStreetMap.

3.3 Use Cases

Table 1 reflects EcoPassEU serves four actor types with differing technical needs. Manufacturers interact with the system through a wallet-connected flow (UC-01 to UC-06), they register product data, upload it to IPFS, and anchor the record on-chain via MetaMask. This design assigns the responsibility for issuing to the party that can best guarantee the accuracy of the product. In contrast, consumers and regulators require no wallet (UC-07 to UC-11), they can search, verify and check any passport just by reading the query, which reflects the principle that transparency should not set up any financial or technical barriers. An admin role (UC-12) supports batch registration, reducing the per-unit cost for high-volume producers.

Table 1. EcoPassEU use cases by actor

ID	Actor	Description
UC-01	Manufacturer	Connect MetaMask wallet (Sepolia testnet)
UC-02	Manufacturer	Complete product registration form
UC-03	Manufacturer	Upload product JSON to IPFS
UC-04	Manufacturer	Register passport on blockchain
UC-05	Manufacturer	Download QR code
UC-06	Manufacturer	View personal product list
UC-07	Consumer/Regulator	Search by product ID (no wallet)
UC-08	Consumer/Regulator	View full passport details
UC-09	Consumer/Regulator	Verify data integrity via hash comparison
UC-10	Consumer/Regulator	View interactive supply chain map
UC-11	Consumer/Regulator	View issuer profile
UC-12	Admin	Batch-register products

4 Project Execution

The implementation of EcoPassEU adopts a hybrid architecture that combines on-chain and off-chain to achieve a balance between data transparency and execution efficiency. The core data structure is distributed in two environments. In

the on-chain part, the smart contract `PassportRegistry.sol` uses a strictly typed struct named `Passport` to store the smallest and immutable data: `ipfsCID` (`string`), `metadataHash` (`bytes32`), issuer’s address and timestamp. The system directly associates the unique product ID with its corresponding data through state mapping, so as to ensure that it can be retrieved quickly. Although the original system design considered using ERC-1155 tokens to represent product passports, we finally chose this lightweight, structure-based registry scheme to significantly reduce the deployment complexity and initial gas cost. In the off-chain part, the complete life cycle data—including product category, brand, material composition and supply chain traceability information—is organised as JSON objects at the front end. The data is uploaded to IPFS through Pinata. Before registering on the chain, the React front-end uses the Ethers.js library to calculate the cryptographic hash value of the JSON.

In order to ensure the stable execution of smart contracts, we performed a complete set of unit tests in the Hardhat environment combined with the Chai assertion library, following established practices in smart contract analysis and validation [14]. The test confirms that the `registerPassport` function can correctly store the issuer’s address—which is automatically derived from the transaction sender to prevent forgery—and generate a valid block timestamp. The test also clearly verified the boundary situation, and successfully ensured that when submitting duplicate product IDs, the transaction will roll back and report errors. In addition, the `batchRegisterPassports` function has also been tested to process multiple arrays at the same time, correctly trigger events for valid entries, and silently skip registered products.

From the perspective of user interaction, the front-end component simplifies the manufacturer’s usage process into a logical and clear operation chain. Manufacturers input materials and traceable data, and use integrated map components to record spatial coordinates. After clicking “Upload to IPFS”, the data will be pinned to IPFS and return a unique content identifier (CID). Finally, clicking “Register on Blockchain” will trigger the user’s MetaMask wallet to sign the transaction and submit the CID and metadata hash to the Sepolia testnet. After successful registration, the system will dynamically generate a QR code mapped to the product URL. Users can directly download the QR code from the interface for physical product labels.

4.1 Critical Evaluation

A critical assessment of the execution cost of the DApp shows that its gas efficiency has been highly optimised. The design of storing larger string data such as repair guides or recycling instructions off-chain ensures that the state changes required for the registry are extremely lightweight. According to the testnet deployment results and standard EVM opcodes estimates, executing a single registration transaction consumes about 125,430 gas. In addition, the `batchRegisterPassports` function shows significant operational scalability. By traversing a set of ID and hash values in one transaction, the contract avoids the basic transaction fee of 21,000 gas. Test data shows that when 10 products

are registered in bulk, the average gas cost of a single product can be reduced to about 92,100 gas, achieving an efficiency improvement of more than 25%, making the platform economically feasible for small and medium-sized enterprises.

In terms of security, the implementation has successfully reduced the risk of tampering and unauthorised modification, which is a central concern in Ethereum smart contract security [17,24]. In smart contracts, the explicit `require` check provides a strict tamperability detection guarantee by making the registered passport immutable. In addition, the stored metadata hash establishes a cryptographic commitment between the on-chain registry and the off-chain IPFS load. If the malicious actor tries to intercept and tamper with the IPFS document, the cryptographic hash of the tampered document will no longer match the immutable record saved on the chain, thus immediately invalidating the passport. Unlike existing solutions that rely on permissioned nodes or proprietary databases, EcoPassEU achieves enterprise-level privacy protection on a fully open network through the IPFS hash pairing mechanism.

Although the decentralised architecture provides strong security, it also brings some runtime restrictions. The DApp performs quickly in on-chain verification and can usually be read in the state of completion in milliseconds. However, the access to the complete product history depends on public IPFS gateways, which sometimes have higher latency than centralised cloud servers. In addition, although the current platform runs efficiently on the Sepolia testnet, if it is migrated to the Ethereum mainnet, users will face volatile gas costs. Future versions of EcoPassEU should be considered for deployment on Layer-2 networks such as Polygon. This can not only maintain EVM compatibility, but also achieve near-instant final confirmation; at the same time, if it is necessary to support secondary market transfer in the future, it can be further reconstructed into a plan based on ERC-1155 tokens [21].

5 Conclusion

Summarize the main theoretical and technical findings. Detail what the group achieved, acknowledge limitations (e.g., the physical-to-digital bridge vulnerabilities), and propose future work.

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A Appendix

This appendix includes the mandatory materials required for the FTGP summative assessment.

A.1 Smart Contract Deployment

- **Smart Contract Address(es):** [Insert Sepolia testnet addresses here]
- **Etherscan Links:** [Insert links to verified contracts on Etherscan]

A.2 Project Links

- **Deployed Website:** [Insert URL]
- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11

A.3 Transaction Hashes

- **Deployment TX:** [Insert Hash] - Explanation.

A.4 Equity Shares & Contributions

- Luxiao Cao: [Weight]
- Akshansh Rajora: [Weight]
- Fuyu Cao: [Weight]
- Tianwei Yu: [Weight]
- Detai Dong: [Weight]

[Insert Screenshot of GitHub - Insights - Contributor here]

A.5 JIRA Project Board

[Insert screenshots of 'Summary', 'timeline', and 'backlog' pages here]

A.6 Individual Reflection Essays

Luxiao Cao [Insert 500-word reflection here]

Akshansh Rajora [Insert 500-word reflection here]

Fuyu Cao [Insert 500-word reflection here]

Tianwei Yu [Insert 500-word reflection here]

Detai Dong [Insert 500-word reflection here]

A.7 Team Sprint Reports

[Insert chronological sprint reports here]

A.8 Complete Source Code

Smart Contracts

```
// Insert complete Solidity code here
```

Frontend Core Files

```
// Insert modified/relevant frontend files (e.g., page.js) here
```